



Department of Computer Science

Program Name : B.Tech.(CS)	Academic Session : 2025-26	Semester:2
Course name : Python for Engineers	Course Code:AI102P	Faculty : Prof. Bhagwan Krishna Gupta

Tagging COs with BLs & KCs			
CO No.	Statement of Course Outcome	Bloom's Cognitive Process Level (BL)	Knowledge Category (KC)
After completion of the course, the student will be able to			
CO1	Understand the basics of Python language and control structures to solve computational problems.	2	C,P
CO2	Demonstrate the Python complex data types (strings, lists, tuples, dictionaries) and functions for efficient data manipulation and problem-solving.	3	C,P
CO3	Apply Object-Oriented Programming concepts and Exception Handling in Python	3	C,P
CO4	Apply Python file operations for reading, writing, manipulating files, and processing structured data efficiently.	3	P
CO5	Develop simple programs utilizing built-in functions of Python packages like Matplotlib, NumPy, and Pandas.	6	P,M

Mapping of Course outcomes with Program outcomes CO-POs Matrix														
Course Name (Course Code)														
Course Code	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12	PSO1	PSO2
CO-1	3	2	1	1	2	1	1	2	2	-	-	-	3	2
CO-2	3	3	2	1	2	1	1	2	2	-	-	-	3	2
CO-3	3	2	3	1	2	1	1	2	2	-	-	-	3	3
CO-4	3	3	2	1	2	1	1	2	2	-	-	-	3	3
CO-5	3	3	3	1	3	2	1	2	2	-	-	-	3	3
PO Target	3	2.6	2.2	1	2.2	1.2	1	2	2	-	-	-	3	3

Signature of Course Coordinator

Signature of Program Head

Signature of Dean

Please Note (Reference: OBE Guidelines wef Session 2023 – 24)

- ❖ The theory courses/ project having credits 3 to 6 should have 5 number of COs. The laboratory course/ mini project/ seminar/ industrial training having credits less than 3 should have 4 number of COs. The Project having 7 to 12 credits should have 6 to 10 number of COs.
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Department of Computer Science

Program Name : B.Tech.(CS)	Academic Session : 2025-26	Semester: 2
Course name : Data Structure	Course Code: CS201B	Faculty : Dr. Rishabh jain

Tagging COs with BLs & KCs			
CO No.	Statement of Course Outcome	Bloom's Cognitive Process Level (BL)	Knowledge Category (KC)
After completion of the course, the student will be able to			
CO1	Use the concept of the array in searching and sorting algorithm.	3	Conceptual, Procedural
CO2	Illustrate the concept of Dynamic Memory Allocation for Operations on Linked List.	4	Conceptual, Procedural
CO3	Analyze different recursion techniques using stack.	4	Conceptual, Procedural
CO4	Implementation of Queue and its applications using basic data structures.	3	Conceptual, Procedural
CO5	Apply the knowledge of tree and binary search tree structures for problem solving.	3	Conceptual, Procedural

Mapping of Course outcomes with Program outcomes CO-POs Matrix														
Course Name (Course Code)														
Course Code	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12	PSO1	PSO2
CO-1	3	2	1	1	2	1	1	2	2	-	-	-	3	2
CO-2	3	3	2	1	2	1	1	2	2	-	-	-	3	2
CO-3	3	2	3	1	2	1	1	2	2	-	-	-	3	3
CO-4	3	3	2	1	2	1	1	2	2	-	-	-	3	3
CO-5	3	3	3	1	3	2	1	2	2	-	-	-	3	3
PO Target	3	2.6	2.2	1	2.2	1.2	1	2	2	-	-	-	3	3

Signature of Course Coordinator

Signature of Program Head

Signature of Dean

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Department of Computer Science

Program Name : B.Tech.(CS)	Academic Session : 2025-26	Semester:2
Course name : Introduction to IoT	Course Code: EE112B	Faculty : Dr. Ameer Faisal

Tagging COs with BLs & KCs			
CO No.	Statement of Course Outcome	Bloom's Cognitive Process Level (BL)	Knowledge Category (KC)
After completion of the course, the student will be able to			
CO1	Understand the basic concepts of sensors and transducers.	Understand	C, P
CO2	Understand basics of embedded system and different IoT boards.	Understand	C, P
CO3	Apply basic operations and programming techniques of IoT devices.	Apply	C, P
CO4	Apply smart technology knowledge through case studies.	Apply	C, P

Mapping of Course outcomes with Program outcomes CO-POs Matrix														
Course Name (Course Code)														
Course Code	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12	PSO1	PSO2
CO-1	2	-	-	-	2	2	2	-	-	-	-	2	-	2
CO-2	2	-	2	-	2	2	2	-	2	-	-	2	-	2
CO-3	3	-	3	2	3	2	2	-	2	-	-	2	-	3
CO-4	3	2	3	3	3	2	2	-	2	-	-	2	-	3
PO Target	2	2	2.66	2.5	2.5	2	2	-	2	-	-	2	-	2.5

Signature of Course Coordinator

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Signature of Dean

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Department of Computer Science

Program Name : B.Tech.(CS)	Academic Session : 2025-26	Semester:2
Course name : Communication Skills	Course Code: HS101P	Faculty : Dr Vipin Tyagi

Tagging COs with BLs & KCs			
CO No.	Statement of Course Outcome	Bloom's Cognitive Process Level (BL)	Knowledge Category (KC)
After completion of the course, the student will be able to			
CO1	Understand the essentials of communicating in a professional setting.	Understand	C
CO2	Employ correct English usage and formal style of Listening - speaking.	Understand	C
CO3	Apply the usage of verbal and non-verbal cues in presentation and day-to-day communication.	Apply	C,P
CO4	Illustrate Communication skills that meet the nature and objectives of the workplace	Apply	C,P

Mapping of Course outcomes with Program outcomes CO-POs Matrix														
Course Name (Course Code)														
Course Code	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12	PSO1	PSO2
CO-1	-	-	-	-	-	-	-	-	2	3	-	1	-	-
CO-2	-	-	-	-	-	-	-	-	2	3	-	1	-	-
CO-3	-	-	-	-	-	-	-	-	2	3	-	1	-	-
CO-4	-	-	-	-	-	-	-	-	2	3	-	1	-	-
PO Target	-	-	-	-	-	-	-	-	2	3	-	1	-	-

Signature of Course Coordinator

Signature of Program Head

Signature of Dean

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Department of Computer Science

Program Name : B.Tech.(CS)	Academic Session : 2025-26	Semester:2
Course name : Linear Algebra for Engineers	Course Code: MA103L	Faculty : Dr Sachin Kumar

Tagging COs with BLs & KCs			
CO No.	Statement of Course Outcome	Bloom's Cognitive Process Level (BL)	Knowledge Category (KC)
After completion of the course, the student will be able to			
CO1	Understand the concept of vector space and subspace.	Understand	C,P
CO2	Apply elementary transformation to solve system of Linear equations.	Apply	C,P
CO3	Understand the concept of complex matrix and Eigen vectors.	Understand	C,P
CO4	Apply the concept of matrix factorization in data decomposition.	Apply	C,P
CO5	Explore the concept of linear transformations and inner product in engineering applications.	Apply	C,P

Mapping of Course outcomes with Program outcomes CO-POs Matrix														
Course Name (Course Code)														
Course Code	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12	PSO1	PSO2
CO-1	3	3	3	-	-	-	-	1	-	-	-	2	-	-
CO-2	2	2	3	-	-	-	-	1	-	-	-	1	-	-
CO-3	2	2	2	-	-	-	-	1	-	-	-	1	-	-
CO-4	2	2	2	-	-	-	-	1	-	-	-	1	-	-
CO-5	2	2	2	-	-	-	-	1	-	-	-	1	-	-
PO Target	2.2	2.2	2.4	-	-	-	-	1	-	-	-	1.2	-	-

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Department of Computer Science

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Program Name : B.Tech.(CS)	Academic Session : 2025-26	Semester:2
Course name : Discrete Structures & Theory of Logic	Course Code: MA202L	Faculty : Dr. Deepti Seth

Tagging COs with BLs & KCs			
CO No.	Statement of Course Outcome	Bloom's Cognitive Process Level (BL)	Knowledge Category (KC)
After completion of the course, the student will be able to			
CO1	Acquire knowledge of sets, relations, and functions to solve problems involving ordered structures and their relationships.	Apply	C,P
CO2	Apply fundamental concepts of lattice and Boolean algebra to strengthen logical reasoning and computational ability.	Apply	C,P
CO3	Employ the rules of propositions, theory of inferences and predicate logic in logical reasoning problems.	Apply	C,P
CO4	Understand the concepts of algebraic structures and their applications to apply in critical thinking.	Apply	C,P
CO5	Apply the concept of graph theory in solving shortest path engineering problems.	Apply	C,P

Mapping of Course outcomes with Program outcomes CO-POs Matrix														
Course Name (Course Code)														
Course Code	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12	PSO1	PSO2
CO-1	2	2	-	-	-	-	-	1	-	-	-	2	-	-
CO-2	2	2	-	2	-	-	-	1	-	-	-	2	-	-
CO-3	2	2	-	2	-	-	-	1	-	-	-	2	-	-
CO-4	2	2	2	2	-	-	-	1	-	-	-	2	-	-
CO-5	2	2	2	2	-	-	-	1	-	-	-	2	-	-
PO Target	2	2	2	2	-	-	-	1	-	-	-	2	-	-

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Department of Computer Science

Program Name : B.Tech.(CS)	Academic Session : 2025-26	Semester:2
Course name : Semiconductor Physics and Devices	Course Code: PH101L	Faculty : Dr. Dharendra Kumar Sharma

Tagging COs with BLs & KCs			
CO No.	Statement of Course Outcome	Bloom's Cognitive Process Level (BL)	Knowledge Category (KC)
After completion of the course, the student will be able to			
CO1	Illustrate the basic concept of crystalline materials and their appropriate use.	Apply	C,P
CO2	Apply the fundamentals of basic semiconductor Physics on transistor and MOSFET.	Apply	C,P
CO3	Apply the concepts of semiconductor Physics in aspect of solar cell and Zener diode.	Apply	C,P
CO4	Implementing of semiconductor Physics to study various characteristics of optoelectronic devices.	Apply	C,P
CO5	Apply the concept of Quantum Physics to study various phenomenon.	Apply	C,P

Mapping of Course outcomes with Program outcomes CO-POs Matrix														
Course Name (Course Code)														
Course Code	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12	PSO1	PSO2
CO-1	2	1	-	-	-	2	2	-	-	2	-	1	-	-
CO-2	3	2	-	-	-	2	2	-	-	2	-	1	-	-
CO-3	3	2	-	-	-	2	2	-	-	2	-	1	-	-
CO-4	3	2	-	-	-	2	2	-	-	2	-	1	-	-
CO-5	2	1	-	-	-	-	-	-	-	1	-	1	-	-
PO Target	2.6	1.6	-	-	-	2	2	-	-	1.8	-	1.0	-	-

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Department of Computer Science

Program Name : B.Tech.(CS)	Academic Session : 2025-26	Semester:2
Course name : Semiconductor Physics and Devices Lab	Course Code: PH101P	Faculty : Dr. Dharendra Kumar Sharma

Tagging COs with BLs & KCs			
CO No.	Statement of Course Outcome	Bloom’s Cognitive Process Level (BL)	Knowledge Category (KC)
After completion of the course, the student will be able to			
CO1	Illustrate the basic concept of crystalline materials and their appropriate use.	Apply	C,P
CO2	Apply the fundamentals of basic semiconductor Physics on transistor and MOSFET.	Apply	C,P
CO3	Apply the concepts of semiconductor Physics in aspect of solar cell and Zener diode.	Apply	C,P
CO4	Implementing of semiconductor Physics to study various characteristics of optoelectronic devices.	Apply	C,P
CO5	Apply the concept of Quantum Physics to study various phenomenon.	Apply	C,P

Mapping of Course outcomes with Program outcomes CO-POs Matrix														
Course Name (Course Code)														
Course Code	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12	PSO1	PSO
CO-1	2	2	-	-	-	-	-	1	2	2	-	1	-	-
CO-2	2	2	-	-	-	-	-	1	2	2	-	1	-	-
CO-3	2	2	-	-	-	-	-	1	2	2	-	1	-	-
CO-4	2	2	-	-	-	-	-	1	2	2	-	1	-	-
CO-5	2	1	-	-	-	-	-	-	1	2	-	1	-	-
PO Target	2	1.8	-	-	-	-	-	1	1.8	2	-	1	-	-

Signature of Course Coordinator

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Department of Computer Science

Program Name : B.Tech.(CS)	Academic Session : 2025-26	Semester: 2nd
Course name : Innovation and Entrepreneurship	Course Code: IT104P	Faculty : Prof. Swati

Tagging COs with BLs & KCs			
CO No.	Statement of Course Outcome	Bloom's Cognitive Process Level (BL)	Knowledge Category (KC)
After completion of the course, the student will be able to			
CO1	Demonstrate an understanding of the various types of innovation, their importance in personal and professional growth, and how to apply innovative thinking to solve real-world problems.	Apply	P
CO2	Gain the ability to generate and refine innovative ideas through creative techniques and utilize the Business Model Canvas to structure viable business concepts.	Create	M
CO3	Develop the skills to conduct comprehensive market research, identify and segment target customers, and validate their business ideas based on market insights and data analysis .	Evaluate	C
CO4	Transform their innovative ideas into tangible prototypes (Minimum Viable Products) and will acquire the ability to craft and deliver compelling pitches for potential investors and stakeholders.	Create	M
CO5	Effectively present their business ideas to industry experts and investors, apply feedback to improve their ideas, and explore opportunities for securing funding or mentorship.	Create	P

Mapping of Course outcomes with Program outcomes CO-POs Matrix														
Innovation and Entrepreneurship (IT104P)														
Course Code	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12	PSO1	PSO
CO-1	2	3	2	1	1	1	1	–	1	–	–	2	–	–
CO-2	1	2	3	1	2	–	–	–	2	1	2	2	–	–
CO-3	1	3	2	3	2	1	–	–	1	1	2	2	–	–
CO-4	1	2	3	2	2	–	–	–	2	3	3	2	2	–
CO-5	–	1	2	1	1	1	–	1	3	3	3	2	–	–
PO Target	1.25	2.2	2.4	1.6	1.6	1	1	1	1.8	2	2.5	2	2	–

Signature of Course Coordinator

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Department of Computer Science

Program Name : B.Tech	Academic Session : 2025-26	Semester: 4th
Course name : Web Technology	Course Code: CS208B	Faculty : Dr. Abhishek Goyal

Tagging COs with BLs & KCs			
CO No.	Statement of Course Outcome	Bloom's Cognitive Process Level (BL)	Knowledge Category (KC)
After completion of the course, the student will be able to			
CO1	Understand advanced ES6 JavaScript features to develop modular and asynchronous applications.	Understand	C,P
CO2	Build cross-platform mobile applications by applying Dart and Flutter with OOP and async concepts.	Apply	C,P
CO3	Design and implement interactive Flutter applications using state management, navigation, and basic animations.	Apply	C,P
CO4	Build web applications by applying Flask and Django, utilizing MVC architecture, form handling, and REST API development.	Apply	C,P

Mapping of Course outcomes with Program outcomes CO-POs Matrix														
Web Technology (CS208B)														
Course Code	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12	PSO1	PSO2
CO-1	3	2	2	-	2	-	-	-	-	-	-	2	2	-
CO-2	3	2	3	-	2	-	-	-	2	-	-	2	2	-
CO-3	2	2	3	-	2	-	-	-	2	1	-	2	2	-
CO-4	3	3	2	2	2	-	-	-	2	1	-	2	2	-
PO Target	2.75	2.25	2.5	2	2	-	-	-	2	1	-	2	2	-

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Department of Computer Science

Program Name : B.Tech	Academic Session : 2025-26	Semester: 4th
Course name : ANN and Machine Learning	Course Code: CS303B	Faculty : Dr. Anurag Mishra

Tagging COs with BLs & KCs			
CO No.	Statement of Course Outcome	Bloom's Cognitive Process Level (BL)	Knowledge Category (KC)
After completion of the course, the student will be able to			
CO1	Preprocess, clean, and transform datasets, and implement supervised ML algorithms (KNN, SVM, Regression) using Python	Apply	C,P
CO2	Apply unsupervised techniques (clustering, PCA) and feature engineering to real-world datasets	Apply	C,P
CO3	Design ensemble models (Random Forest, Stacking) and optimize hyperparameters using AutoML tools	Create	C,P
CO4	Implement ANN models (MLP, RBF) for tasks like digit recognition and music classification	Apply	C,P

Mapping of Course outcomes with Program outcomes CO-POs Matrix														
ANN and Machine Learning (CS303B)														
Course Code	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12	PSO1	PSO2
CO-1	3	3	2	3	3	1	-	-	-	-	-	2	3	-
CO-2	2	3	3	2	3	1	-	-	-	-	-	2	3	-
CO-3	2	3	3	2	3	2	-	-	-	-	-	2	3	-
CO-4	3	2	3	2	3	1	-	-	-	-	-	2	3	-
PO Target	2.5	2.75	2.75	2.25	3	1.25	-	-	-	-	-	2	3	-

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Department of Computer Science

Program Name : B.Tech	Academic Session : 2025-26	Semester: 4th
Course name : Design and Analysis of Algorithms	Course Code: CS401L	Faculty : Prof. Kuldeep Kumar Atariya

Tagging COs with BLs & KCs			
CO No.	Statement of Course Outcome	Bloom's Cognitive Process Level (BL)	Knowledge Category (KC)
After completion of the course, the student will be able to			
CO1	Analyse algorithm efficiency using recurrence relations and sorting techniques.	Analyze	C,P
CO2	Solve optimization problems with Greedy algorithms (MST, Shortest Path).	Apply	C,P
CO3	Implement Dynamic Programming for Knapsack, Matrix Chain Multiplication, etc.	Apply	C,P
CO4	Apply Backtracking and Branch & Bound to combinatorial problems.	Analyze	C,P
CO5	Understand NP-Completeness, Approximation, and Randomized algorithms.	Understand	C,P

Mapping of Course outcomes with Program outcomes CO-POs Matrix														
Design and Analysis of Algorithms (CS401L)														
Course Code	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12	PSO1	PSO2
CO-1	3	3	2	2	2	-	-	1	-	-	1	2	2	3
CO-2	3	3	3	2	2	-	-	1	-	-	1	2	2	3
CO-3	3	3	3	3	2	-	-	1	-	-	1	2	2	3
CO-4	3	3	2	3	2	-	-	1	-	-	1	2	2	3
CO-5	3	3	2	3	2	-	-	1	-	-	1	2	2	2
PO Target	3	3	2.4	2.6	2	-	-	1	-	-	1	2	2	2.8

Signature of Course Coordinator

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Department of Computer Science

Program Name : B.Tech	Academic Session : 2025-26	Semester: 4th
Course name : Universal Human Values	Course Code: HS112L	Faculty : Prof. Aditi Sharma

Tagging COs with BLs & KCs			
CO No.	Statement of Course Outcome	Bloom's Cognitive Process Level (BL)	Knowledge Category (KC)
After completion of the course, the student will be able to			
CO1	Understand the significance of value inputs in a classroom, distinguish between values and skills, understand the need, basic guidelines, content, and process of value education, explore the meaning of happiness and prosperity, and do a correct appraisal of the current scenario in the society	Understand	F,C
CO2	Distinguish between the Self and the Body, and understand the meaning of Harmony in the Self and the Co-existence of Self and Body.	Analyze	F,C
CO3	Understand the value of harmonious relationships based on trust, respect, and other naturally acceptable feelings in human-human relationships and explore their role in ensuring a harmonious society.	Understand	F,C
CO4	Understand the harmony in nature and existence, and workout their mutually fulfilling participation in nature.	Understand	F,C
CO5	Distinguish between ethical and unethical practices, and start working out the strategy to actualize a harmonious environment wherever they work.	Analyze	F,C

Please Note (Reference: OBE Guidelines wef Session 2023 – 24)

- ❖ The theory courses/ project having credits 3 to 6 should have 5 number of COs. The laboratory course/ mini project/ seminar/ industrial training having credits less than 3 should have 4 number of COs. The Project having 7 to 12 credits should have 6 to 10 number of COs.
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Mapping of Course outcomes with Program outcomes CO-POs Matrix														
Universal Human Values (HS112L)														
Course Code	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12	PSO1	PSO2
CO-1	-	-	-	-	-	2	2	2	1	-	1	1	-	-
CO-2	-	-	-	-	-	2	2	2	1	-	1	1	-	-
CO-3	-	-	-	-	-	2	2	2	1	-	1	1	-	-
CO-4	-	-	-	-	-	2	2	2	1	-	1	1	-	-
CO-5	-	-	-	-	-	2	2	3	1	-	1	1	-	-
PO Target	-	-	-	-	-	2	2	2.2	1	-	1	1	-	-

Signature of Course Coordinator

Signature of Program Head

Signature of Dean

Please Note (Reference: OBE Guidelines wef Session 2023 – 24)

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- ❖ The statement of a CO must be formed considering a proper structure having mandatory and optional parts. The mandatory parts are Action & Knowledge and optional parts are Condition and Criteria.



Department of Computer Science

ProgramName :B.Tech	Academic Session: 2025-26	Semester:4
Course Name :Aptitude-2	Course Code:HS113L	Faculty: Prof. Vinod Agrawal

Tagging COs with BLs & KCs			
CO No.	Statement of Course Outcome	Bloom's Cognitive Process Level (BL)	Knowledge Category (KC)
After completion of the course, the student will be able to			
CO1	illustrate their comprehension by solving the given problems.	Understand	C
CO2	apply the learned concepts to new problems and solve them aptly.	Apply	C
CO3	make use of their thought process to interpret and draw inferences from the given data to reach logical conclusions.	Analyse	C

Mapping of Course outcomes with Program outcomes CO-POs Matrix														
Course Name (Course Code)														
Course Code	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12	PSO1	PSO2
CO-1	2	1	-	1	-	1	-	-	-	-	-	1	-	-
CO-2	1	1	-	1	-	2	-	-	-	-	-	1	-	-
CO-3	1	1	-	1	-	1	-	-	-	-	-	2	-	-
PO Target	1.33	1	-	1	-	1.33	-	-	-	-	-	1.33	-	-

Signature of Course Coordinator

Signature of Program Head

Signature of Dean

Please Note (Reference: OBE Guidelines wef Session 2023 – 24)

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- ❖ The statement of a CO must be formed considering a proper structure having mandatory and optional parts. The mandatory parts are Action & Knowledge and optional parts are Condition and Criteria.



Department of Computer Science

Program Name : B.Tech	Academic Session : 2025-26	Semester: 4th
Course name : Soft Skills Essential-2	Course Code: HS114L	Faculty : Dr. Meenu

Tagging COs with BLs & KCs			
CO No.	Statement of Course Outcome	Bloom's Cognitive Process Level (BL)	Knowledge Category (KC)
After completion of the course, the student will be able to			
CO1	Apply advanced communication skills in professional and digital contexts by building a strong online identity, presenting ideas through structured storytelling, and enhancing English language proficiency using digital assessment tools.	Apply	C,P
CO2	Demonstrate professional expression and digital literacy by articulating a structured self introduction through the Driver's Test approach, preparing effective resumes, and writing clear, goal oriented prompts for various LLMs	Apply	C,P
CO3	Exhibit interpersonal effectiveness by navigating negotiation, persuasion, and emotional intelligence in professional contexts.	Apply	C,P

Mapping of Course outcomes with Program outcomes CO-POs Matrix														
Soft Skills Essential-2(HS114L)														
Course Code	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12	PSO1	PSO2
CO-1	-	-	-	-	-	-	-	-	2	3	-	2	-	-
CO-2	-	-	-	-	-	-	-	-	2	3	-	2	-	-
CO-3	-	-	-	-	-	-	-	-	2	3	-	2	-	-
PO Target	-	-	-	-	-	-	-	-	1.6	2.4		1.6	-	-

Signature of Course Coordinator

Signature of Program Head

Signature of Dean

Please Note (Reference: OBE Guidelines wef Session 2023 – 24)

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- ❖ The statement of a CO must be formed considering a proper structure having mandatory and optional parts. The mandatory parts are Action & Knowledge and optional parts are Condition and Criteria.



Department of Computer Science

Program Name : B.Tech	Academic Session : 2025-26	Semester: 4th
Course name : Data Analytics	Course Code: IT202B	Faculty : Prof. Mohit Singh Tanwer

Tagging COs with BLs & KCs			
CO No.	Statement of Course Outcome	Bloom's Cognitive Process Level (BL)	Knowledge Category (KC)
After completion of the course, the student will be able to			
CO1	Explain key concepts, tools, and applications of data analytics.	Understand	C,P
CO2	Apply data cleaning and preprocessing using Python libraries.	Apply	C,P
CO3	Analyze data and visualize insights with BI tools.	Analyze	C,P
CO4	Implement ML algorithms for classification and clustering.	Apply	C,P
CO5	Explore Big data framework and demonstrate Hadoop processing.	Understand	C,P

Mapping of Course outcomes with Program outcomes CO-POs Matrix														
Data Analytics (IT202B)														
Course Code	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12	PSO1	PSO2
CO-1	2	2	1	-	1	-	-	-	-	-	-	1	-	-
CO-2	2	3	2	2	2	-	-	-	-	-	-	2	2	-
CO-3	2	2	3	2	3	1	1	-	1	-	1	2	2	-
CO-4	2	3	3	3	3	1	1	-	1	-	1	2	2	-
CO-5	2	3	3	3	2	1	-	1	1	-	1	2	2	2
PO Target	2	2.6	2.4	2	2.2	1	1	1	1	-	1	1.8	1.6	2

Signature of Course Coordinator

Signature of Program Head

Signature of Dean

Please Note (Reference: OBE Guidelines wef Session 2023 – 24)

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- ❖ The statement of a CO must be formed considering a proper structure having mandatory and optional parts. The mandatory parts are Action & Knowledge and optional parts are Condition and Criteria.



Department of Computer Science

Program Name : B.Tech	Academic Session : 2025-26	Semester: 4th
Course name : Computer Networks	Course Code: IT302L	Faculty : Prof. Pawan Kumar Pal

Tagging COs with BLs & KCs			
CO No.	Statement of Course Outcome	Bloom's Cognitive Process Level (BL)	Knowledge Category (KC)
After completion of the course, the student will be able to			
CO1	Apply the knowledge of networking concepts and functionality of physical layer.	Apply	C,P
CO2	Explore the concept of elementary data link layer protocol to build a robust network	Analyze	C
CO3	Analyze the concept of routing and IP addressing in network layer.	Analyze	C,P
CO4	Examine the usage and working of transport layer.	Apply	C,P
CO5	Determine the performance of different protocols used at application layer	Apply	C

Mapping of Course outcomes with Program outcomes CO-POs Matrix														
Computer Networks (IT302L)														
Course Code	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12	PSO1	PSO2
CO-1	3	2	-	-	-	-	-	-	-	-	-	2	-	-
CO-2	3	3	-	-	2	-	-	-	-	-	-	2	-	-
CO-3	3	3	-	-	2	-	-	-	-	-	2	2	-	-
CO-4	3	3	-	-	-	-	-	-	-	-	-	2	-	-
CO-5	3	3	-	-	-	-	-	-	-	-	-	2	-	-
PO Target	3	2.8	-	-	2	-	-	-	-	-	2	2	-	-

Signature of Course Coordinator

Signature of Program Head

Signature of Dean

Please Note (Reference: OBE Guidelines wef Session 2023 – 24)

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Department of Computer Science

Program Name : B.Tech	Academic Session : 2025-26	Semester:4th
Course name : AWS Foundations & Data Engineering	Course Code: CS335E	Faculty : Prof. Mohit Singh Tanwar

Tagging COs with BLs & KCs			
CO No.	Statement of Course Outcome	Bloom's Cognitive Process Level (BL)	Knowledge Category (KC)
After completion of the course, the student will be able to			
CO1	Understand cloud computing concepts, AWS global infrastructure, and the role of cloud platforms in application and data architectures.	Understand	F,C
CO2	Configure and manage AWS compute, storage, networking, and identity services to support secure and scalable workloads.	Apply	C,P
CO3	Design and implement data ingestion, transformation, and storage pipelines that convert raw data into analytics-ready datasets.	Apply	C,P
CO4	Monitor, optimize, and troubleshoot cloud-based data systems to ensure performance, reliability, and cost efficiency.	Analyze	C,P
CO5	Apply security, governance, and compliance controls to protect data and manage access across cloud-based data platforms.	Apply	C,P

Mapping of Course outcomes with Program outcomes CO-POs Matrix														
AWS Foundations & Data Engineering (CS335E)														
Course Code	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12	PSO1	PSO2
CO-1	3	2	1	-	2	-	-	-	-	-	-	2	2	2
CO-2	3	3	2	2	3	-	-	-	-	-	-	2	3	2
CO-3	3	3	3	2	3	-	-	-	-	-	-	2	3	3
CO-4	3	3	2	3	3	-	-	-	-	-	-	2	3	3
CO-5	3	2	2	2	3	1	1	2	-	-	-	2	3	3
PO Target	3	2.6	2	2.25	2.8	1	1	2				2	2.8	2.6

Signature of Course Coordinator

Signature of Program Head

Signature of Dean

Please Note (Reference: OBE Guidelines wef Session 2023 – 24)

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Department of Computer Science

Program Name : B.Tech	Academic Session : 2025-26	Semester: 4th
Course name : DevOps Foundations & Version Control	Course Code: CS304E	Faculty : Prof. Nishant Raj

Tagging COs with BLs & KCs			
CO No.	Statement of Course Outcome	Bloom's Cognitive Process Level (BL)	Knowledge Category (KC)
After completion of the course, the student will be able to			
CO1	Understand DevOps concepts, agile practices, and the evolution of modern software delivery models.	Understand	C
CO2	Apply Linux fundamentals and system administration skills for DevOps environments.	Apply	C
CO3	Implement distributed version control workflows using Git for collaborative software development.	Apply	C
CO4	Use AI-assisted tools and GitOps practices to enhance code quality, automation, and deployment reliability.	Apply	C
CO5	Apply security, compliance, and productivity analytics practices in DevOps pipelines.	Apply	C

Mapping of Course outcomes with Program outcomes CO-POs Matrix														
DevOps Foundations & Version Control (CS304E)														
Course Code	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12	PSO1	PSO2
CO-1	3	2	1	-	2	-	-	-	1	1	-	2	2	-
CO-2	3	3	2	2	3	-	-	-	1	-	-	2	3	-
CO-3	3	3	3	2	3	-	-	-	2	1	1	2	3	-
CO-4	3	3	3	3	3	-	-	-	2	2	2	2	3	-
CO-5	2	2	2	2	3	1	1	2	2	2	1	3	3	-
PO Target	2.8	2.6	2.2	2.25	2.8	1	1	2	1.6	1.5	1.33	2.2	2.8	-

Signature of Course Coordinator

Signature of Program Head

Signature of Dean

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Department of Computer Science

Program Name : B.Tech	Academic Session : 2025-26	Semester: 4th
Course name : Intelligent Systems with Text & Vision API	Course Code: CS307E	Faculty : Mr. Dheeraj Pandey

Tagging COs with BLs & KCs			
CO No.	Statement of Course Outcome	Bloom's Cognitive Process Level (BL)	Knowledge Category (KC)
After completion of the course, the student will be able to			
CO1	Understand the fundamentals of Natural Language Processing and Computer Vision used in intelligent systems.	Understand	C
CO2	Apply word embeddings, sequence models, and transformer-based approaches for text understanding tasks.	Apply	P
CO3	Implement computer vision pipelines for image processing, object detection, and visual analysis.	Apply	P
CO4	Develop multimodal AI applications by integrating text and vision models using APIs and pre-trained frameworks.	Create	P
CO5	Evaluate, deploy, and optimize intelligent systems while considering performance, ethical, and real-world constraints.	Evaluate	M

Mapping of Course outcomes with Program outcomes CO-POs Matrix														
Intelligent Systems with Text & Vision API (CS307E)														
Course Code	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12	PSO1	PSO2
CO-1	3	2	1	-	2	-	-	-	-	-	-	2	2	-
CO-2	3	3	2	2	3	-	-	-	1	1	-	2	3	-
CO-3	3	3	3	2	3	-	-	-	2	1	1	2	3	-
CO-4	3	3	3	3	3	-	-	-	2	1	2	2	3	-
CO-5	2	2	2	2	3	-	-	-	2	2	2	3	3	-
PO Target	2.8	2.6	2.2	2.25	2.8	-	-	-	1.75	1.25	1.67	2.2	2.8	-

Signature of Course Coordinator

Signature of Program Head

Signature of Dean

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Department of Computer Science

Program Name : B.Tech	Academic Session : 2025-26	Semester: 4th
Course name : Frontend Engineering with React & Next	Course Code: CS318E	Faculty : Prof. Abhishek Sharma

Tagging COs with BLs & KCs			
CO No.	Statement of Course Outcome	Bloom's Cognitive Process Level (BL)	Knowledge Category (KC)
After completion of the course, the student will be able to			
CO1	Understand modern frontend architecture, React fundamentals, and component-based UI development principles.	Understand	F
CO2	Develop dynamic and interactive user interfaces using JSX, state, props, hooks, and reusable components.	Create	P
CO3	Implement scalable state management, routing, and data flow using modern React and Redux patterns.	Create	P
CO4	Build server-rendered and statically generated web applications using Next.js with authentication and API integration.	Create	P
CO5	Deploy, monitor, and optimize frontend applications following industry best practices for performance and scalability.	Evaluate	P

Mapping of Course outcomes with Program outcomes CO-POs Matrix														
Frontend Engineering with React & Next (CS318E)														
Course Code	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12	PSO1	PSO2
CO-1	3	2	1	-	2	-	-	-	-	-	-	2	2	-
CO-2	3	3	2	2	3	-	-	-	1	1	-	2	3	-
CO-3	3	3	3	2	3	-	-	-	2	1	1	2	3	-
CO-4	3	3	2	3	3	-	-	-	2	1	2	2	3	-
CO-5	2	2	2	2	3	-	-	-	2	2	2	3	3	-
PO Target	2.8	2.6	2	1.8	2.8	-	-	-	1.4	1	1	2.2	2.2	-

Signature of Course Coordinator

Signature of Program Head

Signature of Dean

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Department of Computer Science

Program Name : B.Tech	Academic Session : 2025-26	Semester: 4th
Course name : Computer Networks Lab	Course Code: IT302P	Faculty : Prof. Pawan Kumar Pal

Tagging COs with BLs & KCs			
CO No.	Statement of Course Outcome	Bloom's Cognitive Process Level (BL)	Knowledge Category (KC)
After completion of the course, the student will be able to			
CO1	Understand the fundamental concepts of computer networking and Network topologies.	Understand	C
CO2	Analyze different types of network devices and simple computer networks.	Analyze	C,P
CO3	Implement the basic network commands and use techniques, skills, and modern networking tools necessary for engineering practice.	Apply	C,P

Mapping of Course outcomes with Program outcomes CO-POs Matrix														
Computer Networks Lab (IT302P)														
Course Code	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12	PSO1	PSO2
CO-1	2	2	-	-	2	-	-	-	-	-	1	2	-	-
CO-2	3	2	-	-	2	-	-	-	-	-	1	2	-	-
CO-3	3	3	2	2	2	-	-	-	-	2	1	2	-	-
PO Target	2.67	2.33	2	2	2	-	-	-	-	2	1	2	-	-

Signature of Course Coordinator

Signature of Program Head

Signature of Dean

Please Note (Reference: OBE Guidelines wef Session 2023 – 24)

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- ❖ The statement of a CO must be formed considering a proper structure having mandatory and optional parts. The mandatory parts are Action & Knowledge and optional parts are Condition and Criteria.



Department of Computer Science

Program Name : B.Tech	Academic Session : 2025-26	Semester: 4th
Course name : Design and Analysis of Algorithms Lab	Course Code: CS401P	Faculty : Prof. Kuldeep Kumar Atariya

Tagging COs with BLs & KCs			
CO No.	Statement of Course Outcome	Bloom's Cognitive Process Level (BL)	Knowledge Category (KC)
After completion of the course, the student will be able to			
CO1	Implement and analyse Greedy, DP, Backtracking, and Branch & Bound for efficient problem-solving.	Apply	P
CO2	Evaluate algorithm performance and complexity through hands-on programming in sorting, graphs, and optimization.	Apply	P
CO3	Develop problem-solving skills by implementing Knapsack, Matrix Chain Multiplication, Graph Colouring, and TSP.	Apply	P
CO4	Understand NP-Completeness and Approximation Algorithms for tackling computationally hard problems.	Apply	P

Mapping of Course outcomes with Program outcomes CO-POs Matrix														
Design and Analysis of Algorithms Lab (CS401P)														
Course Code	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12	PSO1	PSO2
CO-1	3	3	3	3	2	-	-	-	-	-	-	2	2	-
CO-2	3	3	3	3	3	-	-	-	-	-	-	2	2	-
CO-3	3	3	3	3	3	-	-	-	-	-	-	2	2	-
CO-4	3	3	3	3	2	-	-	-	-	-	-	2	2	-
PO Target	3	3	3	3	2.75	-	-	-	-	-	-	2	2	-

Signature of Course Coordinator

Signature of Program Head

Signature of Dean

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- ❖ The statement of a CO must be formed considering a proper structure having mandatory and optional parts. The mandatory parts are Action & Knowledge and optional parts are Condition and Criteria.



Department of Computer Science

Program Name : B.Tech	Academic Session : 2025-26	Semester: 6th
Course name : Software Engineering	Course Code: BCS601	Faculty : Prof. Shreela Parekh

Tagging COs with BLs & KCs			
CO No.	Statement of Course Outcome	Bloom's Cognitive Process Level (BL)	Knowledge Category (KC)
After completion of the course, the student will be able to			
CO1	Explain various software characteristics and different types of software development models.	Understand	C
CO2	Implement the Software Requirement Specification and basic Software Quality Assurance practices.	Apply	P
CO3	Design software components to meet user requirements using design principles	Apply	P
CO4	Apply automated software testing strategies to the developed software.	Apply	P
CO5	Analyze various software maintenance and project management techniques.	Analyze	P

Mapping of Course outcomes with Program outcomes CO-POs Matrix														
Software Engineering (BCS601)														
Course Code	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12	PSO1	PSO2
CO-1	2	2	2	2	-	-	-	-	-	1	1	2	-	1
CO-2	2	2	2	2	-	1		1	1	2	1	2	-	1
CO-3	3	2	2	2	2	1	-	-	-	1	1	2	-	1
CO-4	3	2	2	2	2	-	-	-	-	1	1	2	-	1
CO-5	2	2	2	2	-	-	-	-	-	1	2	2	-	1
PO Target	2.4	2	2	2	2	1	-	1	1	1.2	1.2	2	-	1

Signature of Course Coordinator

Signature of Program Head

Signature of Dean

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Department of Computer Science

Program Name : B.Tech	Academic Session : 2025-26	Semester: 6th
Course name : Compiler Design	Course Code: BCS602	Faculty : Prof. Arti Sharma

Tagging COs with BLs & KCs			
CO No.	Statement of Course Outcome	Bloom's Cognitive Process Level (BL)	Knowledge Category (KC)
After completion of the course, the student will be able to			
CO1	Acquire basic knowledge of phases and passes of the compiler.	Apply	C, P
CO2	Illustrate Top-Down (LL) and Bottom-up parsers using the YACC tool.	Analyze	C, P, M
CO3	Apply syntax-directed translation method using synthesized and inherited attributes to generate intermediate code.	Apply	C, P
CO4	Analyze data structures used for symbol table, runtime organization and errors in phases of the compiler.	Analyze	C, P
CO5	Apply code optimization and generation techniques for generating target code.	Apply	C, P

Mapping of Course outcomes with Program outcomes CO-POs Matrix														
Compiler Design (BCS602)														
Course Code	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12	PSO1	PSO2
CO-1	3	2	1	1	-	-	-	-	-	-	-	2	-	-
CO-2	3	3	3	2	3	-	-	-	-	-	-	1	-	-
CO-3	3	2	2	1	-	-	-	-	-	-	-	1	-	-
CO-4	3	2	1	2	-	-	-	-	-	-	-	2	-	-
CO-5	3	2	2	3	-	-	-	-	-	-	-	2	-	-
PO Target	3	2.2	1.8	1.8	3	-	-	-	-	-	-	1.6	-	-

Signature of Course Coordinator

Signature of Program Head

Signature of Dean

Please Note (Reference: OBE Guidelines wef Session 2023 – 24)

- ❖ The theory courses/ project having credits 3 to 6 should have 5 number of COs. The laboratory course/ mini project/ seminar/ industrial training having credits less than 3 should have 4 number of COs. The Project having 7 to 12 credits should have 6 to 10 number of COs.
- ❖ The statement of a CO must be formed considering a proper structure having mandatory and optional parts. The mandatory parts are Action & Knowledge and optional parts are Condition and Criteria.



Department of Computer Science

Program Name : B.Tech	Academic Session : 2025-26	Semester: 6th
Course name : Computer Networks	Course Code: BCS603	Faculty : Dr. Sunil Kumar

Tagging COs with BLs & KCs			
CO No.	Statement of Course Outcome	Bloom's Cognitive Process Level (BL)	Knowledge Category (KC)
After completion of the course, the student will be able to			
CO1	Apply the knowledge of networking concepts and functionality of the physical layer.	Understand	C
CO2	Explore the concept of elementary data link layer protocol to build a robust network.	Apply	C,P
CO3	Analyze the concept of routing and IP addressing in the network layer.	Apply	C,P
CO4	Examine the usage and working of the transport layer.	Understand	C
CO5	Determine the performance of different protocols used at the application layer.	Analyze	C,P

Mapping of Course outcomes with Program outcomes CO-POs Matrix														
Computer Networks (BCS603)														
Course Code	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12	PSO1	PSO2
CO-1	3	2	-	-	-	-	-	-	-	-	-	2	-	-
CO-2	3	3	-	-	2	-	-	-	-	-	-	2	1	-
CO-3	3	3	-	-	2	-	-	-	-	-	2	2	1	-
CO-4	3	3	-	-	-	-	-	-	-	-	-	2	-	-
CO-5	3	3	-	-	-	-	-	-	-	-	-	2	-	-
PO Target	3	2.8	-	-	2	-	-	-	-	-	2	2	1	-

Signature of Course Coordinator

Signature of Program Head

Signature of Dean

Please Note (Reference: OBE Guidelines wef Session 2023 – 24)

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- ❖ The statement of a CO must be formed considering a proper structure having mandatory and optional parts. The mandatory parts are Action & Knowledge and optional parts are Condition and Criteria.



Department of Computer Science

Program Name : B.Tech	Academic Session : 2025-26	Semester: 6th
Course name : Essence Of Indian Tradition Knowledge	Course Code: BNC602	Faculty : Prof. Nishant Raj

Tagging COs with BLs & KCs			
CO No.	Statement of Course Outcome	Bloom's Cognitive Process Level (BL)	Knowledge Category (KC)
After completion of the course, the student will be able to			
CO1	Explain the basic features and modalities about Indian states in ancient system.	Understand	C
CO2	Discuss the functioning of Indian literature and culture at the center and state level.	Understand	C
CO3	Differentiate different aspects of Indian religion and philosophies.	Understand	C
CO4	Discuss different laws and regulations related to management and Indian knowledge system.	Understand	C
CO5	Correlate role of engineers with different culture heritage and performing arts.	Understand	C

Mapping of Course outcomes with Program outcomes CO-POs Matrix														
Essence Of Indian Tradition Knowledge (BNC602)														
Course Code	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12	PSO1	PSO2
CO-1	-	-	-	-	-	3	2	2	-	-	-	-	-	-
CO-2	-	-	-	-	-	3	2	2	-	2	-	-	-	-
CO-3	-	-	-	-	-	3	2	3	-	-	-	-	-	-
CO-4	-	-	-	-	-	2	2	2	-	-	-	-	-	-
CO-5	-	-	-	-	-	3	2	2	2	2	-	-	-	-
PO Target	-	-	-	-	-	2.8	2	2.2	2	2	-	-	-	-

Signature of Course Coordinator

Signature of Program Head

Signature of Dean

Please Note (Reference: OBE Guidelines wef Session 2023 – 24)

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Department of Computer Science

Program Name : B.Tech	Academic Session : 2025-26	Semester: 6th
Course name : Big Data	Course Code: BCS061	Faculty : Dr. Gaurav Dubey

Tagging COs with BLs & KCs			
CO No.	Statement of Course Outcome	Bloom's Cognitive Process Level (BL)	Knowledge Category (KC)
After completion of the course, the student will be able to			
CO1	Understand the knowledge of Big Data concepts, its architecture and applications.	Understand	C
CO2	Demonstrate the components of Hadoop and Map Reduce Framework.	Apply	C,P
CO3	Demonstrate the Hadoop Distributed File System and setting up it's environment.	Apply	C,P
CO4	Demonstrate NoSQL database and tools for job scheduling.	Apply	C,P
CO5	Demonstrate Pig, HIVE and HBASE to abstract Hadoop Eco System.	Apply	C,P

Mapping of Course outcomes with Program outcomes CO-POs Matrix														
Big Data (BCS061)														
Course Code	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12	PSO1	PSO2
CO-1	3	2	2	-	-	-	-	-	-	-	2	2	-	2
CO-2	3	3	2	-	2	-	-	-	-	-	2	2	2	2
CO-3	3	1	2	-	-	-	-	-	-	-	2	2	-	2
CO-4	3	3	2	-	3	-	-	-	-	-	2	2	2	2
CO-5	3	3	2	-	3	-	-	-	-	-	2	2	1	2
PO Target	3	2.4	2	1	2.67	-	-	-	-	-	2	2	1.67	2

Signature of Course Coordinator

Signature of Program Head

Signature of Dean

Please Note (Reference: OBE Guidelines wef Session 2023 – 24)

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Department of Computer Science

Program Name : B.Tech	Academic Session : 2025-26	Semester: 6th
Course name : Blockchain Architecture Design	Course Code: BCS063	Faculty : Dr. Aatif Jamshed

Tagging COs with BLs & KCs			
CO No.	Statement of Course Outcome	Bloom's Cognitive Process Level (BL)	Knowledge Category (KC)
After completion of the course, the student will be able to			
CO1	Understand the basics of Blockchain technology along with its primitives.	Understand	C,P
CO2	Explain the requirements for basic protocol along with scalability aspects	Understand	C,P
CO3	Explore the smart contracts using solidity and hyper-ledger fabric tool	Understand	C,P
CO4	Analyze the blockchain techniques for different use cases like Finance, Trade/Supply and Government activities	Analyze	C,P
CO5	Analyze the blockchain techniques for different use cases, such as government activities, to provide security	Analyze	C,P

Mapping of Course outcomes with Program outcomes CO-POs Matrix														
Blockchain Architecture Design (BCS063)														
Course Code	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12	PSO1	PSO2
CO-1	3	1	-	-	1	-	-	-	-	-	-	2	-	1
CO-2	3	2	1	1	1	-	-	-	-	-	-	2	1	2
CO-3	2	2	3	1	3	-	-	-	1	-	-	2	-	1
CO-4	2	3	2	2	1	-	-	-	2	-	1	2	1	2
CO-5	2	3	2	2	1	-	-	-	2	-	1	2	1	3
PO Target	2.4	2.2	2	1.5	1.4	--	-	-	1.67	-	1	2	1	1.8

Signature of Course Coordinator

Signature of Program Head

Signature of Dean

Please Note (Reference: OBE Guidelines wef Session 2023 – 24)

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Department of Computer Science

Program Name : B.Tech	Academic Session : 2025-26	Semester: 6th
Course name : Software Project Management	Course Code: BOE068	Faculty : Prof. Shivani

Tagging COs with BLs & KCs			
CO No.	Statement of Course Outcome	Bloom's Cognitive Process Level (BL)	Knowledge Category (KC)
After completion of the course, the student will be able to			
CO1	Exercise the project planning activities and the key phases of project management.	Apply	P
CO2	Apply various software process models and cost estimation models to develop a project.	Apply	C
CO3	Explore various project activities to compute the critical path for risk analysis.	Analyze	P
CO4	Identify the different project contexts and propose an appropriate management strategy for each.	Analyze	P
CO5	Adapt professional ethics in staff selection and professional concern in team building for successful software development.	Apply	C

Mapping of Course outcomes with Program outcomes CO-POs Matrix														
Software Project Management (BOE068)														
Course Code	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12	PSO1	PSO2
CO-1	3	3	2	1	-	2	-	-	2	3	2	2	-	2
CO-2	2	3	-	-	2	2	-	-	2	-	3	2	-	2
CO-3	3	3	3	3	3	2	-	-	2	2	2	3	-	2
CO-4	2	2	-	2	2	2	-	-	2	2	2	2	-	2
CO-5	1	-	-	-	-	3	2	3	3	3	2	2	-	2
PO Target	2.20	2.75	2.50	2.00	2.33	2.20	2.00	3.00	2.20	2.50	2.20	2.20	-	2

Signature of Course Coordinator

Signature of Program Head

Signature of Dean

Please Note (Reference: OBE Guidelines wef Session 2023 – 24)

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Department of Computer Science

Program Name : B.Tech	Academic Session : 2025-26	Semester: 6th
Course name : Software Engineering Lab	Course Code: BCS651	Faculty : Prof. Shreela Parekh

Tagging COs with BLs & KCs			
CO No.	Statement of Course Outcome	Bloom's Cognitive Process Level (BL)	Knowledge Category (KC)
After completion of the course, the student will be able to			
CO1	Analyze the SRS document for ambiguities, inconsistencies, incompleteness, and its functional and non-functional requirements.	Analyze	P
CO2	Analyze UML use case and class diagrams with appropriate relationships and associations.	Analyze	P
CO3	Examine the logical sequence of activities in a system using suitable pictorial representations.	Analyze	P
CO4	Justify the use of modern engineering tools for software design and testing.	Evaluate	P

Mapping of Course outcomes with Program outcomes CO-POs Matrix														
Software Engineering Lab (BCS651)														
Course Code	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12	PSO1	PSO2
CO-1	2	2	1	2	1	-	-	1	1	1	1	-	1	1
CO-2	2	2	2	2	2	-	-	-	1	1	1	-	2	1
CO-3	2	2	2	2	2	-	-	-	1	1	1	-	2	1
CO-4	2	2	2	2	2	-	-	-	1	1	1	-	2	1
PO Target	2	2	1.75	2	1.75	-	-	1	1	1	1	-	1.75	1

Signature of Course Coordinator

Signature of Program Head

Signature of Dean

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Department of Computer Science

Program Name : B.Tech	Academic Session : 2025-26	Semester: 6th
Course name : Compiler Design Lab	Course Code: BCS652	Faculty : Prof. Arti Sharma

Tagging COs with BLs & KCs			
CO No.	Statement of Course Outcome	Bloom's Cognitive Process Level (BL)	Knowledge Category (KC)
After completion of the course, the student will be able to			
CO1	Implement the Lexical Analyzer using C language and LEX tool.	Apply	C,P
CO2	Experiment with the knowledge of different parsers (Operator precedence, shift reduce etc.) using C language.	Apply	C,P
CO3	Implement Intermediate code generation and optimization for various expressions.	Apply	C,P
CO4	Design a basic tool that showcase phase(s) of the compiler.	Apply	C,M

Mapping of Course outcomes with Program outcomes CO-POs Matrix														
Compiler Design Lab (BCS652)														
Course Code	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12	PSO1	PSO2
CO-1	3	3	-	-	2	-	-	-	1	-	-	2	-	-
CO-2	3	3	-	-	-	-	-	-	1	-	-	2	-	-
CO-3	3	3	-	-	-	-	-	-	1	-	-	2	-	-
CO-4	3	3	1	2	-	-	-	-	2	-	1	2	3	-
PO Target	3	3	1	2	2	-	-	-	1.25	-	1	2	3	-

Signature of Course Coordinator

Signature of Program Head

Signature of Dean

Please Note (Reference: OBE Guidelines wef Session 2023 – 24)

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Department of Computer Science

Program Name : B.Tech	Academic Session : 2025-26	Semester: 6th
Course name : Computer Network Lab	Course Code: BCS653	Faculty : Dr. Sunil Kumar

Tagging COs with BLs & KCs			
CO No.	Statement of Course Outcome	Bloom's Cognitive Process Level (BL)	Knowledge Category (KC)
After completion of the course, the student will be able to			
CO1	Understand the fundamental concepts of computer networking and Network topologies.	Apply	P
CO2	Analyze different types of network devices and simple computer networks	Analyze	C,P
CO3	Implement the basic network commands and use techniques, skills, and modern networking tools necessary for engineering practice.	Evaluate	P

Mapping of Course outcomes with Program outcomes CO-POs Matrix														
Computer Network Lab (BCS653)														
Course Code	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12	PSO1	PSO2
CO-1	2	2	-	-	2	-	-	-	-	-	1	2	1	-
CO-2	3	2	-	-	2	-	-	-	-	-	1	2	1	-
CO-3	3	3	2	2	2	-	-	-	-	2	1	2	1	-
PO Target	2.66	2.33	2	2	2	-	-	-	-	2	1	2	1	-

Signature of Course Coordinator

Signature of Program Head

Signature of Dean

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Department of Computer Science

Program Name : B.Tech	Academic Session : 2025-26	Semester: 8th
Course name : Entrepreneurship Development	Course Code: BOE084	Faculty : Prof. Swati

Tagging COs with BLs & KCs			
CO No.	Statement of Course Outcome	Bloom's Cognitive Process Level (BL)	Knowledge Category (KC)
After completion of the course, the student will be able to			
CO1	Understand the theories of entrepreneurship and Entrepreneurial Development Programmes.	Understand	F
CO2	Understand the theories of entrepreneurship and Entrepreneurial Development Programmes.	Evaluate	C,P
CO3	Understand the importance of Project Management and Project's life cycle	Understand	F,C
CO4	Analyze Project Finance and Project Report.	Analyze	C,P
CO5	Analyze Social Sector Perspectives and Social Entrepreneurship.	Analyze	C,P

Mapping of Course outcomes with Program outcomes CO-POs Matrix														
Entrepreneurship Development (BOE084)														
Course Code	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12	PSO1	PSO2
CO-1	-	-	-	-	-	1	1	2	2	-	3	1	-	-
CO-2	-	-	-	-	-	2	2	3	3	-	3	2	-	-
CO-3	-	-	-	-	-	3	3	2	3	-	3	2	-	-
CO-4	-	-	-	-	-	2	3	2	3	-	3	2	-	-
CO-5	-	-	-	-	-	2	3	3	2	-	3	3	-	-
PO Target	-	-	-	-	-	2	2.4	2.4	2.6	-	3	2	-	-

Signature of Course Coordinator

Signature of Program Head

Signature of Dean

Please Note (Reference: OBE Guidelines wef Session 2023 – 24)

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Department of Computer Science

Program Name : B.Tech	Academic Session : 2025-26	Semester: 8th
Course name : Digital and Social Media Marketing	Course Code: BOE095	Faculty : Prof. Anmol Jain

Tagging COs with BLs & KCs			
CO No.	Statement of Course Outcome	Bloom's Cognitive Process Level (BL)	Knowledge Category (KC)
After completion of the course, the student will be able to			
CO1	Acquire knowledge about Digital Marketing, various channels through which it operates and its role in marketing strategy.	Understand	F, C
CO2	Gain understanding of various social media platforms and the creation of blogs.	Apply	C, P
CO3	Assess the best practices in the digital marketing field across various markets and gain knowledge of various digital marketing tool.	Apply	C
CO4	Formulate Digital marketing Strategies for an organization.	Apply	C, P
CO5	Analyze the privacy, security, content and ethicality issues associated with digital and social media platforms.	Analyze	C

Mapping of Course outcomes with Program outcomes CO-POs Matrix														
Digital and Social Media Marketing (BOE095)														
Course Code	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12	PSO1	PSO2
CO-1	-	-	-	-	-	2	-	-	-	-	-	1	-	1
CO-2	-	-	-	-	-	1	-	-	1	2	-	1	-	-
CO-3	-	-	-	-	2	1	-	-	2	2	-	1	-	1
CO-4	-	-	-	-	2	2	-	-	2	2	-	1	-	1
CO-5	-	-	-	-	-	2	-	-	1	2	-	1	-	-
PO Target	-	-	-	-	2	1.6	-	-	1.5	2		1	-	1

Signature of Course Coordinator

Signature of Program Head

Signature of Dean

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Department of Computer Science

Program Name : B.Tech	Academic Session : 2025-26	Semester: 8th
Course name : Project	Course Code: BCS851	Faculty : Dr. Sunil Kumar

Tagging COs with BLs & KCs			
CO No.	Statement of Course Outcome	Bloom's Cognitive Process Level (BL)	Knowledge Category (KC)
After completion of the course, the student will be able to			
CO1	Identify real-life problems through effective information gathering for comprehensive understanding.	Understand	C
CO2	Apply diverse insights collaboratively to develop a robust conceptual problem-solving model.	Apply	C,P
CO3	Develop a prototype/experimental set-up necessary to complete the project.	Create	C,P
CO4	Examine acquired results thoroughly, drawing meaningful conclusions through in-depth discussion and comprehensive analysis.	Analyze	C,P
CO5	Demonstrate effective teamwork, presentation, and communication skills among team members.	Apply	C,P
CO6	Compile experimental information to publish in journals/conference/Patent.	Create	C,P

Mapping of Course outcomes with Program outcomes CO-POs Matrix														
Project (BCS851)														
Course Code	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12	PSO1	PSO2
CO-1	3	3	3	3	2	-	2	-	3	2	2	3	3	-
CO-2	3	3	3	3	-	-	3	-	-	-	-	3	3	-
CO-3	3	3	3	3	3	-	-	-	-	-	-	3	3	-
CO-4	3	3	3	3	-	-	-	-	-	-	-	3	3	-
CO-5	-	-	-	-	-	-	-	3	3	3	3	3	3	-
CO-6	2	3	-	3	-	-	-	2	-	-	-	3	3	3
PO Target	2.80	3	3	3	2.50	-	2.50	2.50	3	2.50	2.50	3	3	3

Signature of Course Coordinator

Signature of Program Head

Signature of Dean

Please Note (Reference: OBE Guidelines wef Session 2023 – 24)

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